

Q1 - 24 January - Shift 1

Let N denote the number that turns up when a fair die is rolled. If the probability that the system of

equations

$$x + y + z = 1$$

$$2x + Ny + 2z = 2$$

$$3x + 3y + Nz = 3$$

has unique solution is $\frac{k}{6}$, then the sum of value of k

and all possible values of N is

(1) 18

(2) 19

(3) 20

(4) 21

Space for your notes:

Q2 - 24 January - Shift 1

Let Ω be the sample space and $A \subseteq \Omega$ be an event. Given below are two statements :

(S1) : If $P(A) = 0$, then $A = \phi$

(S2) : If $P(A) = 1$, then $A = \Omega$

Then

(1) only (S1) is true

(2) only (S2) is true

(3) both (S1) and (S2) are true

(4) both (S1) and (S2) are false

Space for your notes:

Q3 - 25 January - Shift 1

Let M be the maximum value of the product of two positive integers when their sum is 66. Let the sample space $S = \left\{ x \in \mathbb{Z} : x(66-x) \geq \frac{5}{9}M \right\}$ and the event $A = \{ x \in S : x \text{ is a multiple of } 3 \}$. Then $P(A)$ is equal to

(1) $\frac{15}{44}$

(2) $\frac{1}{3}$

(3) $\frac{1}{5}$

(4) $\frac{7}{22}$

Space for your notes:

Q4 - 25 January - Shift 1

Let x and y be distinct integers where $1 \leq x \leq 25$ and $1 \leq y \leq 25$. Then, the number of ways of choosing x and y , such that $x + y$ is divisible by 5, is _____.

Space for your notes:

Q5 - 25 January - Shift 2

Let N be the sum of the numbers appeared when two fair dice are rolled and let the probability that $N-2, \sqrt{3}N, N+2$ are in geometric progression be $\frac{k}{48}$. Then the value of k is

(1) 2

(2) 4

(3) 16

(4) 8

Space for your notes:

Q6 - 25 January - Shift 2

25% of the population are smokers. A smoker has 27 times more chances to develop lung cancer than a non-smoker. A person is diagnosed with lung cancer and the probability that this person is a smoker is $\frac{k}{10}$. Then the value of k is _____.

Space for your notes:

Q7 - 29 January - Shift 1

Fifteen football players of a club-team are given 15 T-shirts with their names written on the backside. If the players pick up the T-shirts randomly, then the probability that at least 3 players pick the correct T-shirt is

Space for your notes:

- (1) $\frac{5}{24}$ (2) $\frac{2}{15}$
(3) $\frac{1}{6}$ (4) $\frac{5}{36}$

Q8 - 29 January - Shift 2

Let $S = \{w_1, w_2, \dots\}$ be the sample space associated to a random experiment. Let $P(w_n) = \frac{P(w_{n-1})}{2}, n \geq 2$.

Space for your notes:

Let $A = \{2k + 3\ell; k, \ell \in \mathbb{N}\}$ and $B = \{w_n; n \in A\}$.

Then $P(B)$ is equal to

- (1) $\frac{3}{32}$ (2) $\frac{3}{64}$
(3) $\frac{1}{16}$ (4) $\frac{1}{32}$

Q9 - 30 January - Shift 1

If an unbiased die, marked with $-2, -1, 0, 1, 2, 3$ on its faces, is through five times, then the probability that the product of the outcomes is positive, is :

(1) $\frac{881}{2592}$

(2) $\frac{521}{2592}$

(3) $\frac{440}{2592}$

(4) $\frac{27}{288}$

Space for your notes:

Q10 - 30 January - Shift 2

A bag contains six balls of different colours. Two balls are drawn in succession with replacement.

The probability that both the balls are of the same colour is p . Next four balls are drawn in succession with replacement and the probability that exactly three balls are of the same colours is q . If $p : q = m : n$, where m and n are coprime, then $m + n$ is equal to _____.

Space for your notes:

Q11 - 31 January - Shift 1

A bag contains 6 balls. Two balls are drawn from it at random and both are found to be black. The probability that the bag contains at least 5 black balls is

(1) $\frac{5}{7}$

(2) $\frac{2}{7}$

(3) $\frac{3}{7}$

(4) $\frac{5}{6}$

Space for your notes:

Q12 - 31 January - Shift 2

Let A be the event that the absolute difference between two randomly chosen real numbers in the sample space $[0, 60]$ is less than or equal to a

. If $P(A) = \frac{11}{36}$, then a is equal to _____.

Space for your notes:

Q13 - 01 February - Shift 1

In a binomial distribution $B(n, p)$, the sum and product of the mean & variance are 5 and 6 respectively, then find $6(n + p - q)$ is equal to :-

(1) 51

(2) 52

(3) 53

(4) 50

Space for your notes:

Q14 - 01 February - Shift 2

Two dice are thrown independently. Let A be the event that the number appeared on the 1st die is less than the number appeared on the 2nd die, B be the event that the number appeared on the 1st die is even and that on the second die is odd, and C be the event that the number appeared on the 1st die is odd and that on the 2nd is even. Then

- (1) the number of favourable cases of the event $(A \cup B) \cap C$ is 6
- (2) A and B are mutually exclusive
- (3) The number of favourable cases of the events A, B and C are 15, 6 and 6 respectively
- (4) B and C are independent

Space for your notes:

Answer Key

Q13 (2) **Q14 (1)**

#MathBoleTohMathonGo

To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

Q1 (3)

$$x + y + z = 1$$

$$2x + Ny + 2z = 2$$

$$3x + 3y + Nz = 3$$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & N & 2 \\ 3 & 3 & N \end{vmatrix}$$

$$= (N - 2)(N - 3)$$

For unique solution $\Delta \neq 0$

$$\text{So } N \neq 2, 3$$

$$\Rightarrow P(\text{system has unique solution}) = \frac{4}{6}$$

$$\text{So } k = 4$$

$$\text{Therefore sum} = 4 + 1 + 4 + 5 + 6 = 20$$

Q2 (4)

Ω = sample space

A = be an event

$$\text{If } P(A) = 0 \Rightarrow A = \phi$$

$$\text{If } P(A) = 1 \Rightarrow A = \Omega$$

Then both statement are true

Q3 (2)

$$M = 33 \times 33$$

$$x(66 - x) \geq \frac{5}{9} \times 33 \times 33$$

$$11 \leq x \leq 55$$

$$A : \{12, 15, 18, \dots, 54\}$$

$$P(A) = \frac{15}{45} = \frac{1}{3}$$

Q4 (120)

$$x + y = 5\lambda$$

Cases :

x	y	Number of ways
5λ	5λ	20
$5\lambda + 1$	$5\lambda + 4$	25
$5\lambda + 2$	$5\lambda + 3$	25
$5\lambda + 3$	$5\lambda + 2$	25
$5\lambda + 4$	$5\lambda + 1$	25

Q5 (2)

$$n(s) = 36$$

Given : $N - 2, \sqrt{3N}, N + 2$ are in G.P.

$$3N = (N - 2)(N + 2)$$

$$3N = N^2 - 4$$

$$\Rightarrow N^2 - 3N - 4 = 0$$

$$(N - 4)(N + 1) = 0 \Rightarrow \boxed{N = 4} \text{ or } N = -1 \text{ rejected}$$

$$(\text{Sum} = 4) \equiv \{(1, 3), (3, 1), (2, 2)\}$$

$$n(A) = 3$$

$$P(A) = \frac{3}{36} = \frac{1}{12} = \frac{4}{48} \Rightarrow \boxed{k = 4}$$

Q6 (9)

E_1 : Smokers

$$P(E_1) = \frac{1}{4}$$

E_2 : non-smokers

$$P(E_2) = \frac{3}{4}$$

E : diagnosed with lung cancer

$$P(E/E_1) = \frac{27}{28}$$

$$P(E/E_2) = \frac{1}{28}$$

$$P(E_1 / E) = \frac{P(E_1)P(E / E_1)}{P(E)}$$

$$= \frac{\frac{1}{4} \times \frac{27}{28}}{\frac{1}{4} \times \frac{27}{28} + \frac{3}{4} \times \frac{1}{28}} = \frac{\cancel{27}^9}{\cancel{30}_{10}} = \frac{9}{10}$$

$$K = 9$$

Q7 (3)

#MathBoleTohMathonGo

Required probability =

$$1 - \frac{D_{(15)} + {}^{15}C_1 \cdot D_{(14)} + {}^{15}C_2 \cdot D_{(13)}}{15!}$$

Taking $D_{(15)}$ as $\frac{15!}{e}$ $D_{(14)}$ as $\frac{14!}{e}$ $D_{(13)}$ as $\frac{13!}{e}$

$$\text{We get, } 1 - \left(\frac{\frac{15!}{e} + 15 \cdot \frac{14!}{e} + \frac{15 \times 14}{2} \times \frac{13!}{e}}{15!} \right)$$

$$= 1 - \left(\frac{1}{e} + \frac{1}{e} + \frac{1}{2e} \right) = 1 - \frac{5}{2e} \approx .08$$

Q8 (2)

$$\text{Let } P(w_1) = \lambda \text{ then } P(w_2) = \frac{\lambda}{2} \dots P(w_n) = \frac{\lambda}{2^{n-1}}$$

$$\text{As } \sum_{k=1}^{\infty} P(w_k) = 1 \Rightarrow \frac{\lambda}{1 - \frac{1}{2}} = 1 \Rightarrow \lambda = \frac{1}{2}$$

$$\text{So, } P(w_n) = \frac{1}{2^n}$$

$$A = \{2k + 3\ell; k, \ell \in \mathbb{N}\} = \{5, 7, 8, 9, 10, \dots\}$$

$$B = \{w_n : n \in A\}$$

$$B = \{w_5, w_7, w_8, w_9, w_{10}, w_{11}, \dots\}$$

$$A = \mathbb{N} - \{1, 2, 3, 4, 6\}$$

$$\therefore P(B) = 1 - [P(w_1) + P(w_2) + P(w_3) + P(w_4) + P(w_6)]$$

$$= 1 - \left[\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} \right]$$

$$= 1 - \frac{32 + 16 + 8 + 4 + 1}{64} = \frac{3}{64}$$

Q9 (2)

Either all outcomes are positive or any two are negative.

$$\text{Now, } p = P(\text{positive}) = \frac{3}{6} = \frac{1}{2}$$

$$q = p(\text{negative}) = \frac{2}{6} = \frac{1}{3}$$

Required probability

$$= {}^5C_5 \left(\frac{1}{2}\right)^5 + {}^5C_2 \left(\frac{1}{3}\right)^2 \left(\frac{1}{2}\right)^3 + {}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{1}{2}\right)^1$$

$$= \frac{521}{2592}$$

$\therefore$ Option (2) is correct.

Q10 (14)

$$p = \frac{{}^6C_1}{6 \times 6} = \frac{1}{6}$$

$$q = \frac{{}^6C_1 \times {}^5C_1 \times 4}{6 \times 6 \times 6 \times 6} = \frac{5}{54}$$

$$\therefore p : q = 9 : 5 \Rightarrow m + n = 14$$

Q11 (1)

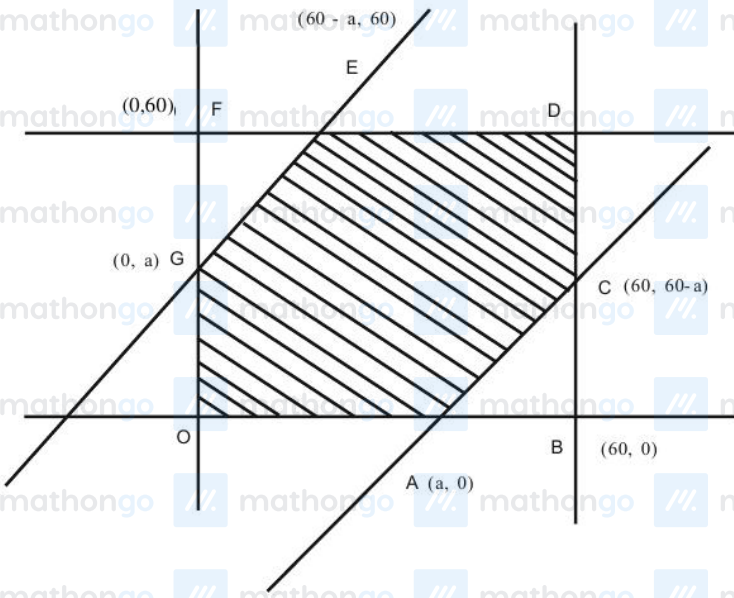
$$\frac{{}^5C_2 + {}^6C_2}{{}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^8C_2} = \frac{10 + 15}{1 + 3 + 6 + 10 + 15}$$

$$= \frac{25}{35} = \frac{5}{7}$$

Q12 (10)

$$|x - y| < a \Rightarrow -a < x - y < a$$

$$\Rightarrow x - y < a \text{ and } x - y > -a$$



$$P(A) = \frac{\text{ar}(\text{OACDEG})}{(\text{OBDF})}$$

$$= \frac{\text{ar}(\text{OBDF}) - \text{ar}(\text{ABC}) - \text{ar}(\text{EFG})}{\text{ar}(\text{OBDF})}$$

$$\Rightarrow \frac{11}{36} = \frac{(60)^2 - \frac{1}{2}(60-a)^2 - \frac{1}{2}(60-a)^2}{3600}$$

$$\Rightarrow 1100 = 3600 - (60-a)^2$$

$$\Rightarrow (60-a)^2 = 2500 \Rightarrow 60-a = 50$$

$$\Rightarrow a = 10$$

Q13 (2)

$$np + npq = 5, np \cdot npq = 6$$

$$np(1 + q) = 5, n^2 p^2 q = 6$$

$$n^2 p^2 (1 + q)^2 = 25, n^2 p^2 q = 6$$

$$\frac{6}{q} (1 + q)^2 = 25$$

$$6q^2 + 12q + 6 = 25q$$

$$6q^2 - 13q + 6 = 0$$

$$6q^2 - 9q - 4q + 6 = 0$$

$$(3q - 2)(2q - 3) = 0$$

$$q = \frac{2}{3}, \frac{3}{2}, q = \frac{2}{3} \text{ is accepted}$$

$$p = \frac{1}{3} \Rightarrow n \cdot \frac{1}{3} + n \cdot \frac{1}{3} \cdot \frac{2}{3} = 5$$

$$\frac{3n + 2n}{9} = 5$$

$$n = 9$$

$$\text{So } 6(n + p - q) = 6 \left(9 + \frac{1}{3} - \frac{2}{3} \right) = 52$$

Q14 (1)

A : no. on 1st die < no. on 2nd die

A : no. on 1st die = even & no. of 2nd die = odd

C : no. on 1st die = odd & no. on 2nd die = even

$$n(A) = 5 + 4 + 3 + 2 + 1 = 15$$

$$n(B) = 9$$

$$n(C) = 9$$

$$n((A \cup B) \cap C) = (A \cap C) \cup (B \cap C) \\ = (3 + 2 + 1) + 0 = 6.$$